

HAT-P-36b

R-0794

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_250412 · created 2026-07-28T08:07:05+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460777.6693 +/- 0.0057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.034
Transit depth (Rp/Rs)^2	2.39 +/- 1.04 [%]
Orbital Inclination (inc)	80.84 +/- 2.74
Airmass coefficient 1 (a1)	1.102 +/- 0.023
Airmass coefficient 2 (a2)	-0.087 +/- 0.046
Scatter in the residuals of the lightcurve fit is	2.11 %
Best Comparison Star	#2 - [217, 342]
Optimal Method	PSF photometry
Transit Duration (day)	0.071 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.155 ± 0.034 vs 0.126 (deviation 0.85σ)
Duration fit vs published	0.071 d vs 0.0925 d (deviation 1.43σ)
Mid-time O-C	-5.2 min
Depth SNR	4.6
Residual scatter	2.11 %

Light curve

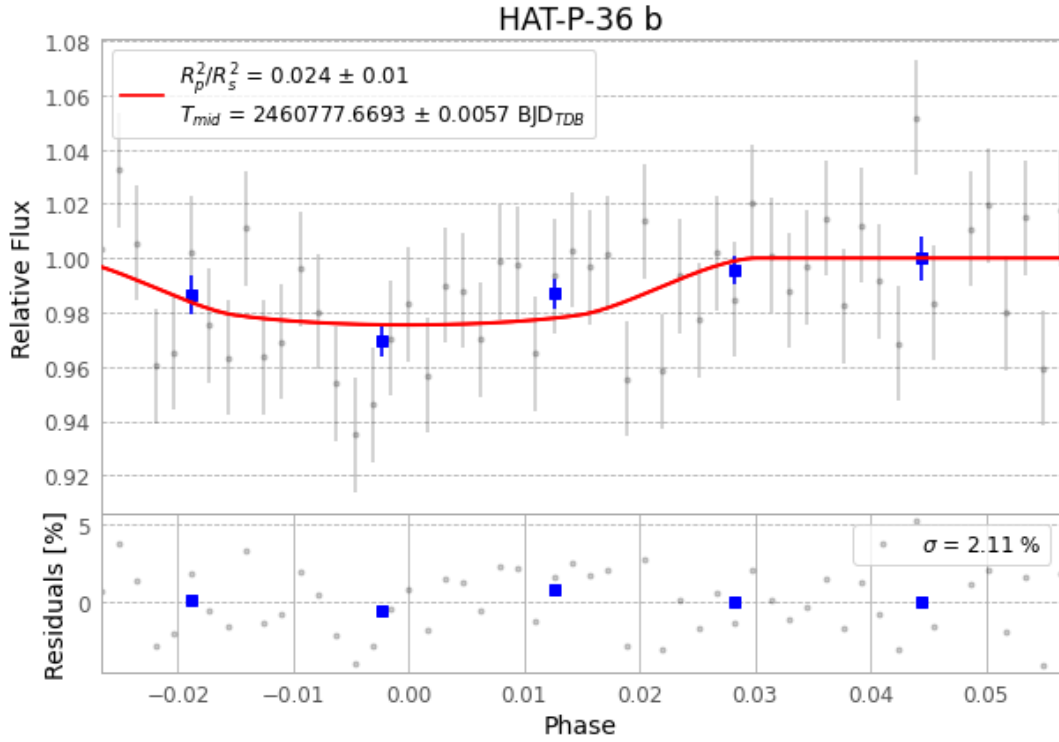


Figure 1 — FinalLightCurve_HAT-P-36 b_2025-04-11.png (SHA-256 fab3adec9df45b29...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [361, 197], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8704f9a89ed37f3a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

54 FITS frames (35 MB), HATP-36250412030616.FITS → HATP-36250412054517.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250412.log (f4380707b665...), FinalLightCurve_HAT-P-36 b_2025-04-11.png (fab3adec9df4...), FinalLightCurve_HAT-P-36 b_2025-04-11.csv (d48a0892c17b...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 08:07Z.